Auteur: Rune Suy
Studierichting: Informatica
Oefeningenreeks 3G nr. 11.4

$$f(x) = \frac{3x^2 - 12x + 9}{x - 2}$$

1. Domein van $f(x)$:

$\text{dom}(f) = \mathbb{R}\backslash\{2\}$

2. Asymptoten:

$$\lim_{x\to 2-} f(x) = \lim_{x\to 2-} \frac{-3}{0} = +\infty$$
$$\lim_{x\to 2+} f(x) = \lim_{x\to 2-} \frac{-3}{0} = -\infty$$

Er is dus een verticale asymptoot in $x = 2$. Langs links gaat f naar $+\infty$,

langs rechts naar $-\infty$

$$\lim_{x\to\infty} \frac{f(x)}{x} = \lim_{x\to\infty} \frac{3x^2 + 12x + 9}{x^2 - 2x} = 3 = a$$
$$\lim_{x\to\infty} f(x) - ax = \lim_{x\to\infty} \frac{\cancel{3x^2} - 12x + 9 - \cancel{3x^2} + 6x}{x - 2} = \lim_{x\to\infty} \frac{-6x}{x} = -6$$

SA: $y = 3x - 6$

geen HA

$$f(x) - 3x + 6 = \frac{\cancel{3x^2} - \cancel{12x} + 9 - \cancel{3x^2} + \cancel{6x} + \cancel{6x} - 12}{x - 2} = \frac{-3}{x - 2}$$

TO:

x		2	
T	$-$	$-$	$-$
N	$-$	0	$+$
$f(x) - 3x + 6$	$+$	$\vert$	$-$

dus f ligt boven SA bij $-\infty$ en onder SA bij $+\infty$

3. Tekenonderzoek van $f(x)$:

$D = 144 - 4 \cdot 3 \cdot 9 = 36$

$$x_1 = \frac{12 + 6}{6} = 3$$
$$x_2 = \frac{12 - 6}{6} = 1$$

TO:

x		1		2		3	
T	$+$	0	$-$	$-$	$-$	0	$+$
N	$-$	$-$	$-$	0	$+$	$+$	$+$
$f(x)-3x+6$	$-$	0	$+$	$\vert$	$-$	0	$+$

4. Tekenonderzoek $f'(x)$:

$$f'(x) = d\left(\frac{3x^2-12x+9}{x-2}\right) = \frac{(6x-12)(x-2)-(3x^2-12x+9)}{(x-2)^2}$$
$$= \frac{6x^2-12x-12x+24-3x^2+12-9}{(x-2)^2} = \frac{3x^2-12x+15}{(x-2)^2}$$

T: $D = 144 - 4 \cdot 3 \cdot 15 = -36$

x		2	
T	$-$	$-$	$-$
N	$-$	0	$+$
$f'(x)$	$+$	$\vert^{(2)}$	$-$
$f(x)$	$\nearrow$	$\vert$	$\nearrow$

5. Tekenonderzoek $f''(x)$:

$$f''(x) = d\left(\frac{3x^2-12x+15}{(x-2)^2}\right) = \frac{(6x-12)(x^2-4x+4)-(2x-4)(3x^2-12+15)}{(x-2)^4}$$
$$= \frac{-6x+12}{(x-2)^4} = \frac{-6(x-2)}{(x-2)^4} = \frac{-6}{(x-2)^3}$$

TO:

x		2	
T	$-$	$-$	$-$
N	$-$	0	$+$
$f''(x)$	$+$	$\vert^{(3)}$	$-$
$f(x)$	$\smile$	$\vert$	$\frown$

6. Schets:

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References